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SETU: A Selective Triage and Correct Framework for Hallucination Mitigation in Bengali Small Language Models

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INTRODUCTION

The **SETU** (Selective **E**valuation, **T**riage, and **U**ncertainty) framework named after the Bengali word for "*bridge*" to reflect how it bridges detection and mitigation addresses this challenge:

Small Language Models (SLMs) frequently produce fluent yet factually inaccurate or ungrounded outputs (hallucinations). Although Bengali is the sixth most spoken language globally [1] , there remains a critical shortage of dedicated frameworks to mitigate these issues.

MOTIVATION

Bengali

- 6th most spoken language worldwide
- 300+ million speakers
- Low-resource Language

Small Models

- 1–3B models hallucinate significantly more than large models
- Most deployable option for low-resource settings like Bangla Language

Why this matters: Bengali speakers are a huge population with almost no dedicated safety tooling for their language, small models are what's actually affordable to deploy in this setting, and critically nobody has built a framework that fixes hallucinations instead of just flagging them. That gap is exactly what SETU sets out to close.

PROBLEM STATEMENT

Bengali SLMs frequently produce fluent but hallucinated outputs, yet existing tools only detect errors instead of fixing them. Moreover, current methods treat all mistakes uniformly without separating factual errors from logical errors. SETU solves this by providing the first selective triage and active correction framework for Bengali SLMs

OBJECTIVES

- To identify the most reliable uncertainty signal for detecting hallucination in Bengali.
- To classify hallucinations as either data-driven or reasoning-driven.
- To correct data-driven errors using cross-lingual retrieval-augmented generation (RAG).
- To correct reasoning-driven errors using Chain-of-Verification (CoVe).
- To enable the framework to honestly abstain when a correction cannot be verified.
- To evaluate the framework across three tasks: QA, Summarization, and Reasoning.
- To validate the contribution of each component through an ablation study.

LITERATURE REVIEW

No	Name of the Paper	Authors	Publisher & Year	Observation
01	HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models[2]	Li et al.	EMNLP, 2023	Method: : Two-stage sampling-then-filtering pipeline ChatGPT generates hallucinated samples, then filters for the most plausible/difficult ones. Result: ChatGPT hallucinates in ~19.5% of responses; only 62.59% accuracy detecting QA hallucinations, 58.53% in summarization. Findings: LLMs struggle to detect hallucinations; external knowledge helps a lot, CoT is mixed, ground-truth contrast hurts performance. Dataset: 35,000 samples 5,000 general queries (Alpaca) + 30,000 task-specific (QA/HotpotQA, dialogue/OpenDialKG, summarization/CNN-DailyMail).
02	HalluSearch: A Search-Enhanced RAG Pipeline for Hallucination Detection [3]	Raúl Vázquez et al.	SemEval, 2025	Method: Splits output into atomic (single-fact) claims, retrieves web/Wikipedia evidence per claim, and verifies via NLI entailment/contradiction with a confidence fallback. Result: F1 ~0.73 on English (top systems); atomic splitting improved F1 from 0.61 → 0.73 (+12%). Findings: Low-resource languages dropped to F1 ~0.54 (–19% vs. English); bad retrieval can be worse than no retrieval (F1 0.58). Dataset: Mu-SHROOM benchmark :14 languages; 250 validation + 150 test items per language (10 languages with full splits) + live Wikipedia/web retrieval as evidence.

LITERATURE REVIEW(CONT.)

03	SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models [4]	Manakul et al.	EMNLP, 2023.	Method: Generates multiple samples per answer and measures inter-sample agreement (self-consistency) as a zero-resource, black-box uncertainty signal. Result: AUC on WikiBio — NLI variant 0.830, BERT-score 0.801, n-gram 0.762, vs. perplexity baseline 0.714. Findings: NLI-based consistency is the strongest variant and needs no external resources or model internals. Dataset: WikiBio: 238 sampled Wikipedia biography passages → 1,908 annotated sentences.
04	Detecting Hallucinations in Large Language Models Using Semantic Entropy [5]	Farquhar et al.	Nature, 2024.	Method: Clusters multiple sampled answers by meaning (not exact wording) via NLI, then measures the entropy of meaning-clusters as an uncertainty signal. Result: AUC-ROC 0.879 (TriviaQA), 0.847 (SQuAD), 0.831 (BioASQ) best reported; beats token-level entropy by +8.3% and exact-match consistency by +11.2%. Findings: Meaning-level clustering captures hallucination better than surface-level or token-level measures. Dataset: TriviaQA ~95,000 QA pairs, SQuAD ~100,000+ QA pairs, BioASQ ~4,721 expert-curated biomedical QA pairs.

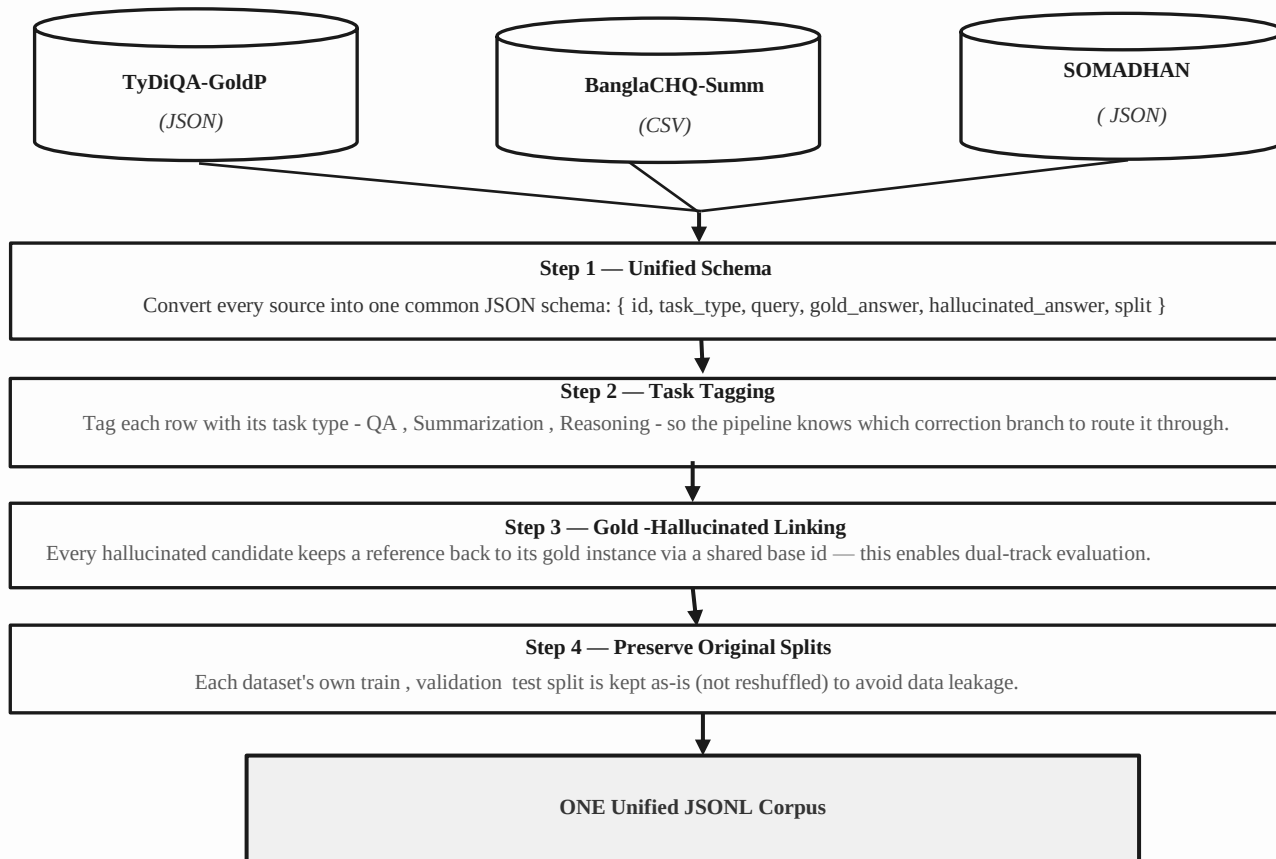
LITERATURE REVIEW(CONT.)

05	Chain-of-Verification Reduces Hallucination in Large Language Models [6]	Dhuliawala et al.	ACL, 2024.	Method:The model answers a question, then double-checks itself by asking follow-up questions. It answers those follow-up questions separately, then uses them to fix its first answer. Result: Got 13% more accurate on one task (58% → 71%) Got better at multi-part answers too (0.39 → 0.48) Made way fewer wrong facts in biography writing (2.95 → 0.68) Findings: It works well, but it's slow takes 3-4x more time/compute since it has to ask and answer extra questions. Never tested on Bengali. Biggest problem: it treats every mistake the same way, no matter if it's a fact error or a logic error. Dataset: WikiBio: Wikidata questions small set, ~600 items QUEST list 55 questions MultiSpanQA 418 questions Biography writing no fixed dataset, just asks the model to write and checks the facts after
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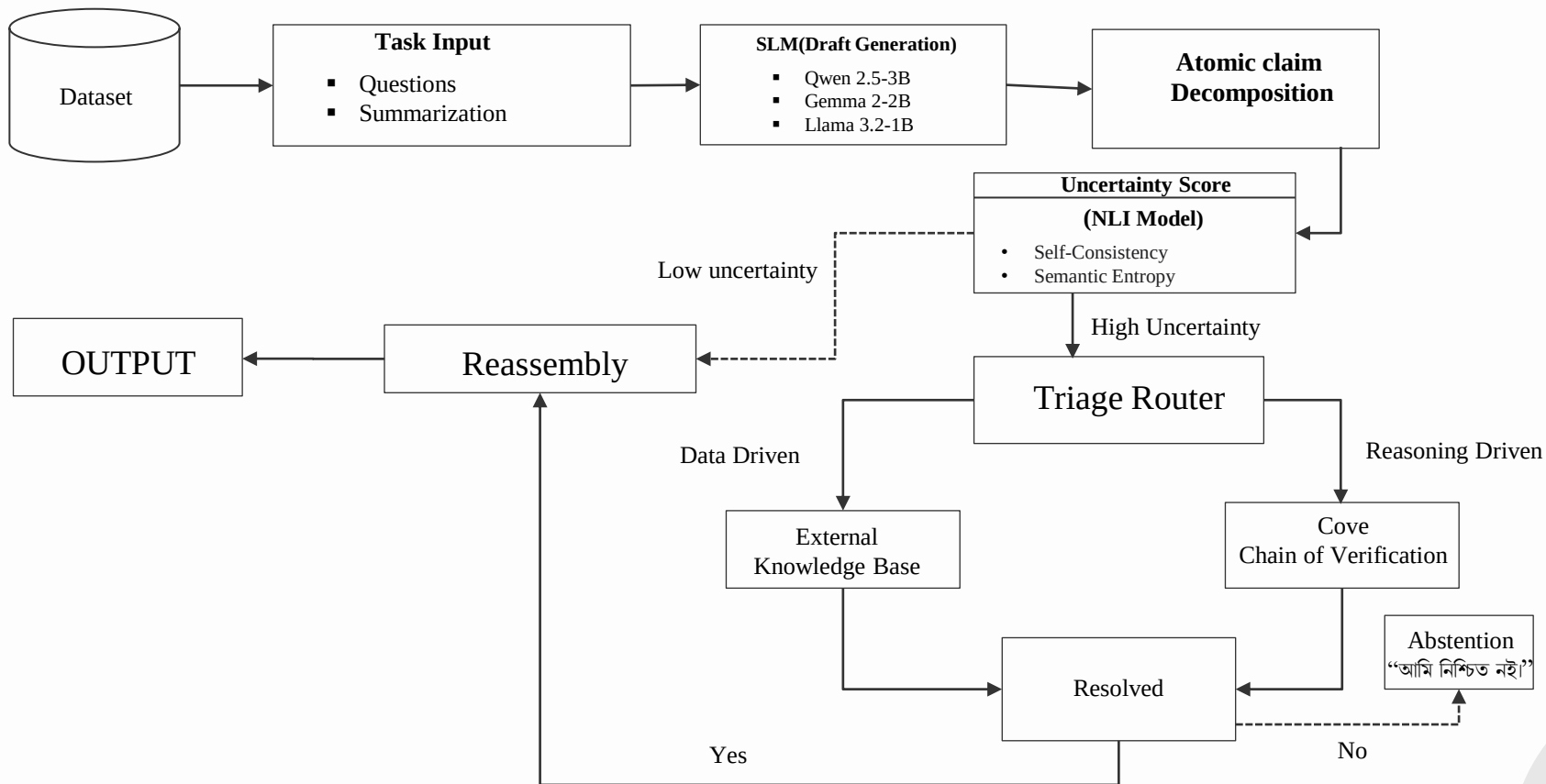
DATASET

Task Category	Dataset Source	Instances	Source
Generative QA	TyDiQA-GoldP (Bengali)	5,000 (1K Gold + 4K Hallu.)	https://huggingface.co/datasets/google-research-datasets/tydiqa
Summarization	BanglaCHQ-Summ	4,000 (1K Gold + 3K Hallu.)	https://github.com/alvi-khan/BanglaCHQ-Summ
Reasoning	SOMADHAN	2,000 (1K Gold + 1K Hallu.)	https://data.mendeley.com/datasets/34bs5cxk9j/1
Detection Val.	BenHalluEval	8,000 total	https://arxiv.org/abs/2605.31483

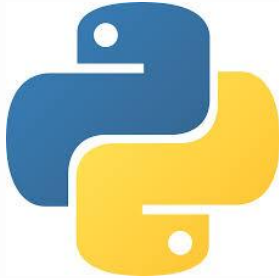
DATASET MERGING STRATEGY



PROPOSED METHODOLOGY



REQUIRED TOOLS



Python



Google Colab



Jupyter Notebook

CONCLUSION

- SETU is the first framework to actively fix Bengali hallucinations instead of just identifying them.
- The project provides the first study on which uncertainty signals work best for native Bengali and code-mixed inputs.
- The system forces models to admit they are unsure when a claim cannot be verified to prevent misinformation.
- Dual-track evaluation ensures the model catches actual errors without creating too many false alarms.
- The framework is optimized for Small Language Models to ensure reliable performance on affordable hardware.

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